CMMI Level 5 Process Architecture

Unified Funeral Services Management Platform

Continuous Process Improvement (FSD-29)

Causal Analysis & Resolution

Defect Detection → Root Cause Analysis (5 Whys / Fishbone) → Corrective Action Implementation → Preventive Action Implementation → Knowledge Repository Update

Innovation Management

Idea Generation → Proof of Concept → Pilot Program → Scale Deployment → Innovation ROI Tracking

Quantitative Process Management

Process Performance Baselines → Statistical Process Control → Process Capability Assessment → Process Performance Modeling → Continuous Process Improvement

Quantitative Project Management (FSD-30)

Statistical Process Control

Control Charts (X-bar / R) → Process Behavior Analysis → Outlier Detection → Trend Analysis → Process Adjustment

Predictive Modeling

Schedule Predictability Models → Quality Predictability Models → Cost Predictability Models → Risk Prediction Models → Resource Forecasting Models

Project Performance Management

Milestone Tracking → Earned Value Management → Defect Density Analysis → Productivity Metrics → Customer Satisfaction Tracking

CMMI Maturity Levels

Level 1 — Initial

- Ad hoc
- Unpredictable
- Reactive

Level 2 — Managed

- Basic project management
- Requirements management
- Planning
- Monitoring
- Supplier agreements

Level 3 — Defined

- Standard processes defined
- Organizational process
- Integrated teams
- Training programs
- Peer reviews

Level 4 — Quantitatively Managed

- Statistical process control
- Quantitative management
- Predictive models
- Process performance baselines

Level 5 — Optimizing

- Continuous process improvement
- Innovation management
- Causal analysis
- Defect prevention
- Technology change management